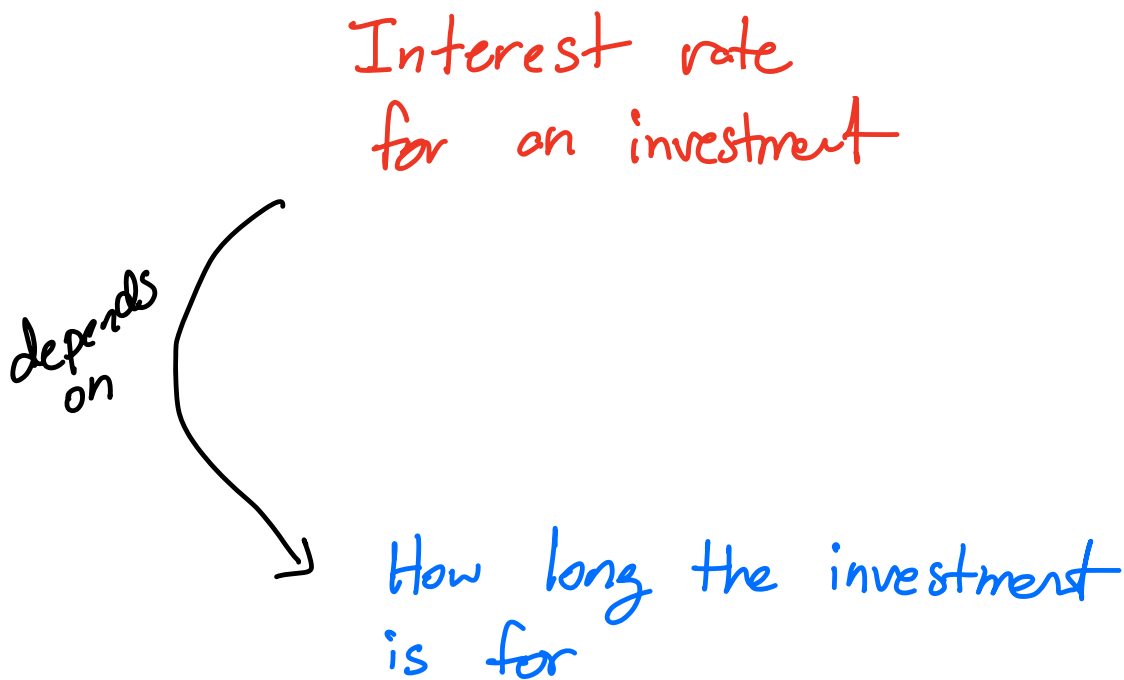


8.3 The term structure of interest rates

Throughout this course in any investment scenario if we saw that $i = 5\%$ we have thought of this global fact that the effective interest rate is 5% from this point on.

Sometimes we have seen changes in interest rates. But we haven't really seen the following fact:



This idea is known as the

term structure
of interest rates

Varying Term Structure Theories

1. (Pure) Expectations Theory

A sequence of short term bonds = A single long-term bond

If y_n is the yield rate of an n -year investment

$1 + y_n$ is the geometric mean of a

sequence of 1-year bonds each at
time $t \in \{1, 2, \dots, n\}$

$$1 + y_n = \left(\prod_{t=1}^n (1 + f_{[t-1, t]}) \right)^{1/n}$$

To believe this theory to explain why yield rates on long term bonds are greater than short term bonds you'd have to believe: Interest rates are increasing

2. Liquidity Preference Theory (opportunity Cost Theory)

A sequence of short
term bonds

>

A single
long-term bond

\$ more easily accessible

To overcome this preference issuers of long-term bonds must offer higher yield rates or a premium known as a liquidity Premium

3. Preferred Habitat Theory

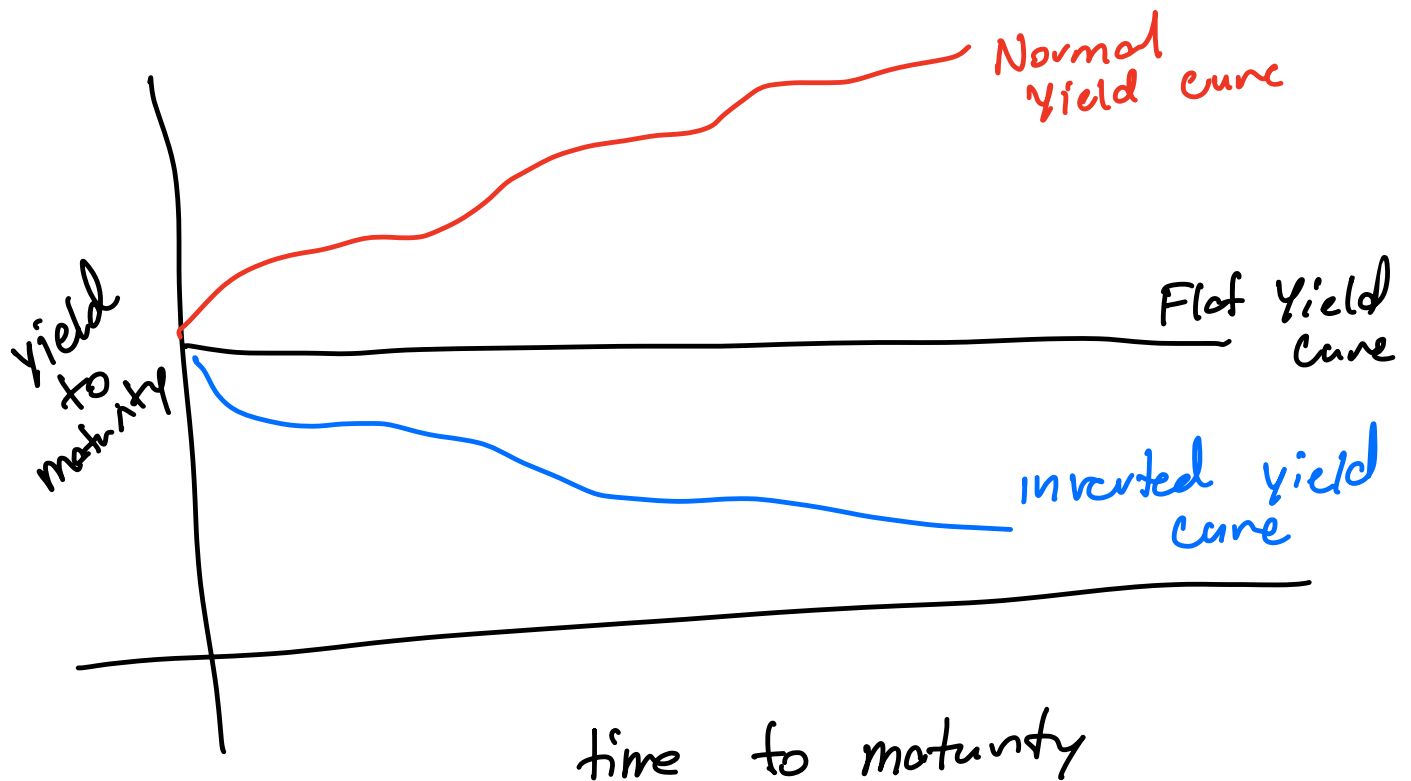
every investor has their own preference and may be induced to leave their preferred habitat if compensated properly. This difference is called a Term Premium

4. Market Segmentation Theory

The bond market is not one market but rather segmented and across segments, yield rates are independent.

The Yield Curve is a way to visualize the investors yield rate for a set of bonds with various times to maturity.

Historically yield curve often refers to the U.S. Treasury Bonds which have (zero default) risk



definition. For $t > 0$ the spot rate r_t is the annual effective rate earned by money invested now for t years

$$(1 + r_t)^t = 1 + i_{[0,t]}$$

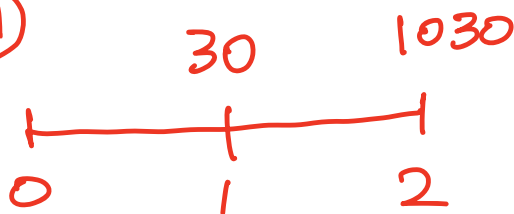
Problem: A two-year bond pays annual coupons of \$30 and matures at \$1,000. Its price is \$984. The same corporation sells a one-year \$1,000 zero-coupon bond for \$975. Compute the two-year spot rate r_2 .

Bond 1

Solution. $N=2$ $m=1$, $F=1000$, $C=30$, $P=984$

Bond 2 (zero coupon) $N=1$, $F=1000$, $P=975$

(B1)



$$984 = \frac{30}{(1+r_1)} + \frac{1030}{(1+r_2)^2}$$

(B2)



$$975 = \frac{1000}{(1+r_1)}$$

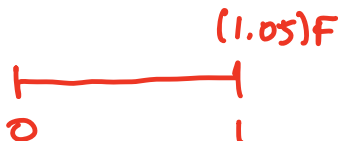
so $r_1 = 2.564\%$ and $r_2 = 3.866\%$

<u>Term</u>	<u>Yield</u>
1 Year	$y_1 = 1.7568\%$
2 Year	$y_2 = 3.0153\%$
3 Year	$y_3 = 3.5463\%$
4 Year	$y_4 = 3.8616\%$
5 Year	$y_5 = 4.2984\%$

TABLE 8.3.4 (5% par-value bonds with annual coupons)

Problem: Use Table (8.3.4) to determine the spot rate r_5 . Note that the bonds are all 5% bonds with annual coupons.

solution.



$$y_1 = r_1$$



$$P = .05F(1+y_2)^{-1} + 1.05F(1+y_2)^{-2}$$

$$P = .05F(1+r_1)^{-1} + 1.05F(1+r_2)^{-2}$$

$$F \left(\frac{.05}{1.030153} + \frac{1.05}{1.030153^2} \right) = F \left(\frac{.05}{1.017568} + \frac{1.05}{(1+r_2)^2} \right)$$

$$\Rightarrow r_2 = 3.04656379\%$$

$$F \left(\frac{.05}{1+y_3} + \frac{.05}{(1+y_3)^2} + \frac{1.05}{(1+y_3)^3} \right) = F \left(\frac{.05}{1+r_1} + \frac{.05}{(1+r_2)^2} + \frac{1.05}{(1+r_3)^3} \right)$$

$$r_3 = 3.593881343\%$$

Same method yields $r_4 = 3.921504795\%$
 $r_5 = 4.392870872\%$

Notice to find r_5 we found $\{r_k\}_{k=1}^5$
This is called bootstrapping.

Notice $r_t > y_t$ for all $t > 1$.

9.1 Interest rate sensitivity Overview

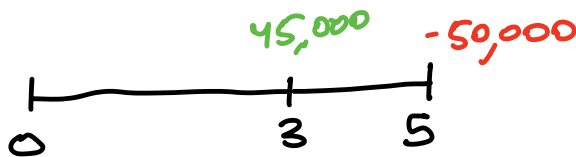
Imagine you owe 50,000 in five years



to cover this liability, you could just buy a 5-year zero-coupon bond and call it a day. But what if 3-year or 7-year bond prices were more attractive?

Scenario 1: 3-year zero coupon bond for same price as 5-year zero coupon bond.

In this case the 3-year bond will mature for less than 50,000 (maybe think 45,000 FV) so in your



head you intrinsically believe that money will grow to 50,000 under some interest rate assumption.

What could go wrong:

Lower interest rates than expected

Scenario 2: You purchase a 7-year zero-coupon bond and then sell it at time 5 to obtain 50,000\$ but since the price of this resale depends on the interest rate at time 5, the lower the interest rate, the larger the price... But

What could go wrong:

Higher interest rates than expected

Idea of Immunization: Can we split our investment between the scenarios to mitigate any risk

Remember we could have just purchased the 5-yr bond and been done: Asset Matching
But sometimes immunization is a favorable investment.

Problem: As in the preceding paragraphs, suppose that you need to pay \$50,000 in five years and that you can finance this with zero-coupon bonds with terms of three years and seven years. Imagine that you buy a three-year zero-coupon bond with a \$22,675.74 redemption amount and a seven-year zero-coupon bond redeemable at maturity for \$27,562.51, each priced to yield 5% annually. Suppose also that, at the end of three years, no matter what the yield rate i (from a flat yield curve) may be, you sell the remaining bond at a purchase price to yield i , combine the proceeds with the \$22,675.74 from the redeemed bond, and use the total to buy a two-year zero-coupon bond. Illustrate that this immunizes against interest rate risk by showing that it produces the needed \$50,000 five years after your initial bond purchases if $i = 20\%$ (a high rate), $i = 5\%$ (the present moderate rate), or $i = 1\%$ (a low rate).

Solution.

3-yr Bond: $N=3, C=22,675.74$ } $j=5\%$
 7-yr Bond: $N=7, C=27,562.51$

Case 1: $i = 20\%$

$$\text{Value at time 3} \rightarrow \left(22,675.74 + \frac{27,562.51}{(1.2)^4} \right) (1.2)^2 = 51,793.70$$

Case 2: $i = 5\%$

$$\text{Value at time 3} \rightarrow \left(22,675.4 + \frac{27,562.51}{(1.05)^4} \right) (1.05)^2 = 50,000$$

Case 3: $i = 1\%$

$$\text{value at time 3} \rightarrow \left(22,675.4 + \frac{27,562.51}{(1.01)^4} \right) (1.01)^2 = 50,150.6$$

We now want to measure: How changes in interest rates change the value of cashflows

Fix a set of cash flows and let $P(i)$ be the P.V. given interest rate i .

↑
price
curve

If all cash flows are positive then $P(i)$ decreases as $i \rightarrow \infty$

The duration measures how sensitive $P(i)$ is to a shift in interest rates

If cash flows have duration $\bar{t}$ $\Rightarrow$ for small h it will lose about $\bar{t}h$ of its value if the interest rate increases by h ($i_0 \rightarrow i_0 + h$) and it will go up in value $\bar{t}h$ if interest rates decrease by h .

Note: duration is just an approximation since the yield curve may not be flat. But it is more accurate the smaller h is.

There will be many types of duration the above is called

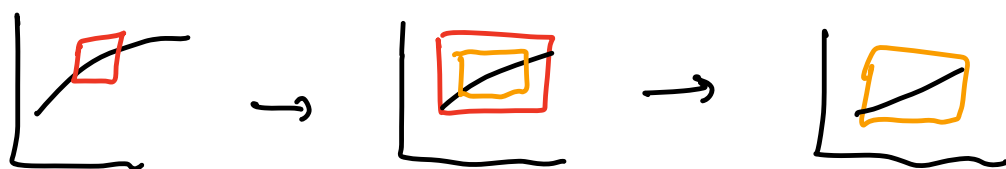
Modified Duration: $D(i_0, 1)$

Another type is Macaulay Duration

$$D(i_0, \infty) = (1 + i_0) D(i_0, 1)$$

↳ conceptually: Average length of an investment

Modified duration doesn't capture the way the price curve is shaped since for small h we always see a linear function



$P(i_0+h) \approx P(i_0)(1-Dh)$ when portfolio has duration D .

Convexity gives us more detail on the shape of the price curve.

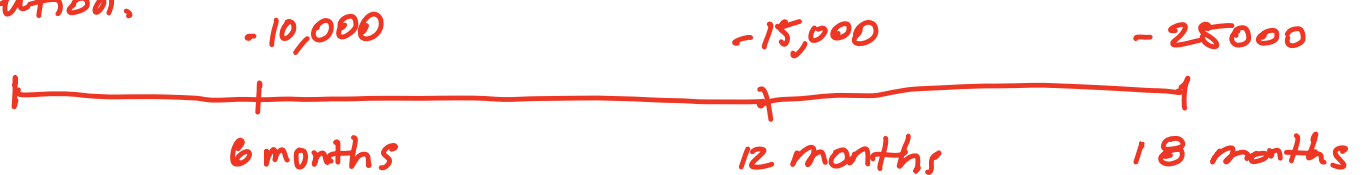
Example of Asset-Liability Matching

Problem: Sandy is obligated to pay \$10,000 in six months, \$15,000 in twelve months, and \$25,000 in eighteen months. He wishes to purchase bonds to exactly match these liabilities — that is, to provide inflows so that the net cashflow at the three payment times, as well as at all other times, is zero. The bonds available for purchase by Sandy are of the following three types:

- (a) Six-month zero-coupon bonds, sold to yield the investor 6% nominal interest convertible semiannually;
- (b) Twelve-month 6% par-value bonds with semiannual coupons;
- (c) Eighteen-month 5% par-value bonds with semiannual coupons.

How much of each of these should Sandy purchase? Assume that each may be purchased for any par value that Sandy would like.

Solution.



start w/ furthest time: let F_C be face amount for bond C.

earned $\Rightarrow .025 F_C + 1.025 F_C$ setting $1.025 F_C = 25000$

we get $F_C = 24,390.24$

$\Rightarrow$ Coupon is $609.76 \$$ Thus, @ $t=12$ months

$609.76 + 1.03 F_B = 15000 \Rightarrow F_B = 13971.1068$

$\Rightarrow$ Coupon is 419.13

Thus, @ $t=6$ months

$609.76 + 419.13 + F_A = 10,000 \Rightarrow F_A = 8971.11$

9.2 Duration

In this section we make two assumptions

1. Flat term structure of interest rates
(In other words, all spot rates are equal)
2. If one spot rate changes, all spot rates change by the same amount.
("parallel shifts")

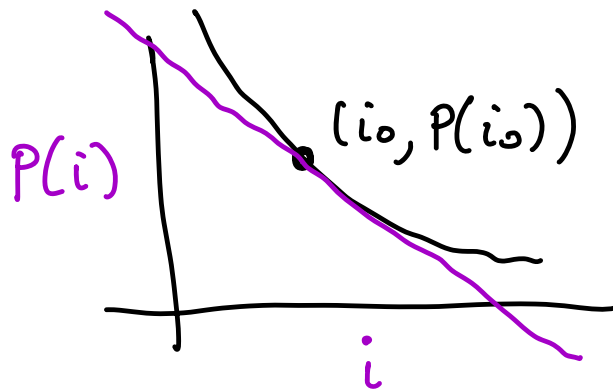
Goal of this section:

Understand how the value of cash flows shift as interest rates shift

Given cash flows $\{C_t : t \geq 0\}$

$$P(i) = \sum_{t \geq 0} C_t (1+i)^{-t}$$

If i_0 is an initial interest rate, and $P(i_0) > 0$ we may have



tangent line: slope is $P'(i_0)$ so

$$P(i) - P(i_0) \approx P'(i_0)(i - i_0)$$

More Taylor series terms $\Rightarrow$ better approx.

ex:

$$P(i) \approx P(i_0) + P'(i_0)(i - i_0) + \frac{P''(i_0)}{2}(i - i_0)^2$$

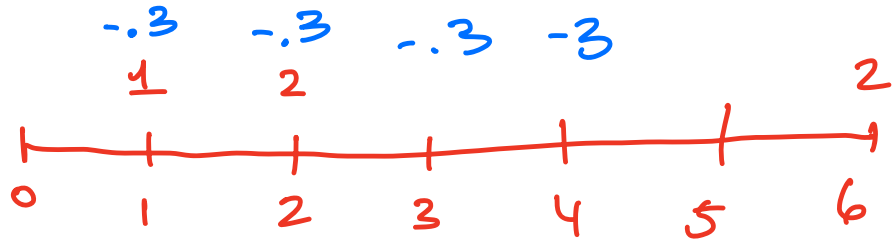
what are the derivatives?

$$P'(i) = -\sum_{t \geq 0} C_t \cdot t(1+i)^{-t-1}$$

$$P''(i) = \sum_{t \geq 0} C_t \cdot t(t+1)(1+i)^{-t-2}$$

Problem: Excel Finance holds loans that guarantee they receive repayments of \$1,000,000 in one year, \$2,000,000 in two years, and \$2,000,000 in six years. They also have issued bonds that require them to make payments of \$300,000 at the end of each of the next three years and \$3,000,000 in four years. Calculate Excel Finance's price function at 2%, 4%, 4.9%, 5%, 5.1%, 6%, and 10%. Then use the tangent line approximation at $i_0 = 5\%$ to estimate the prices at each of these rates, excluding 5%. Also use the second Taylor polynomial around $i_0 = 5\%$ to estimate the prices at 2%, 4%, 6%, and 10%. Discuss your findings.

Solution.



Actual PV's

$$P(2\%) = .7(1.02)^{-1} + 1.7(1.02)^{-2} - .3(1.02)^{-3} - 3(1.02)^{-4} + 2(1.02)^{-6} = 1.0419$$

$$P(4\%) = .99434$$

$$P(4.9) = .97575573$$

$$P(5) = .9737887$$

$$P(5.1) = .97184087$$

$$P(6) = .95512561$$

$$P(10) = .89583537$$

$$P'(5\%) =$$

Tan line approx. $P(i) = P(5\%) + P'(5\%)(i - 5\%)$

$$P(2\%) = 1.03251$$

$$P(4\%) = .99336285$$

$$P(4.9\%) = .97574627$$

$$P(5.1) = .97183147$$

$$P(6) = .95421489$$

$$P(10) = .87591896$$

} Really Good Approx!!!

$$P(i) - P(i_0) \approx P'(i_0)(i - i_0) \leftarrow \begin{matrix} \text{tangent} \\ \text{approx} \end{matrix}$$

So

$$\frac{P(i) - P(i_0)}{P(i_0)} \approx \frac{P'(i_0)}{P(i_0)} (i - i_0)$$

This gives us: "Fractional Price Change"
(relative)

terminology: If rates go from $q\%$
to $(q+1)\%$ we say they
increased by one hundred
basis points.

Problem: Use Approximation (9.2.9) to estimate the relative price change in Example (9.2.8) when i changes from .05 to .051.

Solution. $P'(5\%) = -1.9573982$, $P(5\%) = .9737887$

$$\text{Price Change} \approx \frac{P'(5\%)}{P(5\%)} (.01) = -.020100851$$

The term $P'(i_0)/P(i_0)$ plays a key role in rate change approximations

definition. Modified Duration

$$D(i, 1) = \frac{-P'(i)}{P(i)}$$

The 1 in $D(i, 1)$ comes from using an effective interest rate (nominal 1 time) per year

More generally, we have

$$D(i, m) = - \frac{\frac{dP}{di^{(m)}}}{P(i)}$$

$$\text{where } 1+i = \left(1 + \frac{i^{(m)}}{m}\right)^m$$

by chain rule

$$\frac{dP}{di^{(m)}} = \frac{dP}{di} \cdot \frac{di}{di^{(m)}} = P'(i) \left(\frac{1+i}{1 + \frac{i^{(m)}}{m}} \right)$$

Thus,

$$D(i, m) = D(i, 1) \left(\frac{1+i}{1 + \frac{i^{(m)}}{m}} \right)$$

a quick manipulation shows

$$D(i, m) \left(1 + \frac{i^{(m)}}{m}\right) = D(i+1) (1+i)$$

we also define

$$D(i, \infty) = \frac{-\frac{dP}{d\delta}}{P(i)}$$

$$\text{where } \delta = \ln(1+i)$$

$$e^{\delta} - 1 = i$$

$$\text{so } \frac{di}{d\delta} = e^{\delta}$$

$$\text{so } D(i, \infty) = D(i+1) (1+i)$$

equivalently

$$D(i, \infty) = D(i, m) \left(1 + \frac{i^{(m)}}{m}\right) \text{ for any } m$$

$$D(i, \infty) = D(i, 1) (1+i)$$

$$D(i, \infty) = \lim_{m \rightarrow \infty} D(i, m)$$

we call $D(i, \infty)$ Macaulay Duration

$$\text{Modified duration} = \text{Macaulay Duration} / (1+i)$$

$$D(i, \infty) = \frac{-\frac{dP}{ds}}{P(i)} = \frac{\sum_{t \geq 0} C_t \cdot t e^{-st}}{\sum_{t \geq 0} C_t e^{-st}}$$

$$= \sum_{t \geq 0} \left(\frac{C_t (1+i)^{-t}}{P(i)} \right) t$$

Think: Macaulay duration is a weighted average of the pmt times with weights

$$\left(C_t (1+i)^{-t} / P(i) \right) \leftarrow \text{proportion of PV. that cash flow } C_t \text{ contributes!}$$

Problem: An N -year zero-coupon bond is purchased to provide an annual effective yield i . Find the Macaulay duration $D(i, \infty)$ and the modified duration $D(i, 1)$.

Solution.

$$D(i, 1) = \frac{-P'(i)}{P(i)} \quad \text{where } P(i) = C(1+i)^{-N}$$
$$\text{so } P'(i) = -CN(1+i)^{-N-1}$$

$$D(i, 1) = \frac{CN(1+i)^{-N-1}}{C(1+i)^{-N}} = \frac{N}{(1+i)} \Rightarrow D(i, \infty) = N.$$

To see $D(i, \infty)$

$$P(s) = Ce^{-sN} \Rightarrow P'(s) = -CNe^{-sN}$$

$$\text{So } D(i, \infty) = \frac{-P'(s)}{P(s)} = N.$$

Problem: Find the Macaulay duration $D([(1.03)^2 - 1], \infty)$ of a ten-year 8% \$15,000 bond with semiannual coupons and redemption amount \$16,500.

Solution. $i = 1.03^2 - 1 = .0609$

$$\begin{array}{llll} N=10 & \alpha=8\% & C=16,500 & 1+i = (1+j)^2 \\ m=2 & r=4\% & j=3\% & j = (1+i)^{1/2} - 1 \\ n=20 & F=15000 & & \end{array}$$

$$\begin{aligned} P(i) &= 600 a_{\overline{20}|j} + 16,500(1+j)^{-20} \\ &= 600 \left(\frac{1 - (1+j)^{-20}}{j} \right) + 16,500(1+j)^{-20} \end{aligned}$$

$$= 600 \left(\frac{1 - (1+i)^{-10}}{(1+i)^{1/2} - 1} \right) + 16,500(1+i)^{-10}$$

$$\begin{aligned} P'(i) &= 600 \left(\frac{10(1+i)^{-11}((1+i)^{1/2} - 1) - (1 - (1+i)^{-10})\left(\frac{1}{2}(1+i)^{-1/2}\right)}{\left((1+i)^{1/2} - 1\right)^2} \right) \\ &\quad - 10(16,500)(1+i)^{-11} \end{aligned}$$

$$\begin{aligned} \frac{-P'(.0609)}{P(.0609)} &= D(i, 1) \Rightarrow D(i, \infty) = (1.0609)(\text{ans}) \\ &= 7.4110475 \end{aligned}$$

9.3 Convexity

Just like how $\frac{P'(i)}{P(i)}$ was a main character

in estimating $P(i)$ and led to the study of duration, $\frac{P''(i)}{P(i)}$ can also be used

in estimating $P(i)$ (second order Taylor Series)
so we define

Modified Convexity

$$C(i, 1) = \frac{P''(i)}{P(i)}$$

and similarly

Macaulay Convexity

$$C(i, \infty) = \frac{P''(s)}{P(s)}$$

since $P(s) = \sum_{t \geq 0} C_t e^{-st}$

$$P'(s) = \sum_{t \geq 0} -t C_t e^{-st} \quad \text{and} \quad P''(s) = \sum_{t \geq 0} t^2 C_t e^{-st}$$

$$\text{so } C(i, \infty) = \frac{\sum_{t \geq 0} t^2 C_t e^{-st}}{\sum_{t \geq 0} C_t e^{-st}}$$

$$= \frac{t_1^2 C_{t_1} e^{-st_1} + t_2^2 C_{t_2} e^{-st_2} + \dots}{C_{t_1} e^{-st_1} + C_{t_2} e^{-st_2} + \dots}$$

$$= \sum_{t \geq 0} \frac{C_t e^{-st}}{\sum_{t \geq 0} C_t e^{-st}} t^2$$

$$= t_1^2 \left(\frac{C_{t_1} e^{-st_1}}{C_{t_1} e^{-st_1} + C_{t_2} e^{-st_2} + \dots} \right) + t_2^2 \left(\frac{C_{t_2} e^{-st_2}}{C_{t_1} e^{-st_1} + C_{t_2} e^{-st_2} + \dots} \right) + \dots$$

$$= \sum_{t \geq 0} \left(\frac{C_t (1+i)^{-t}}{P(i)} \right) t^2$$

So again we see the convexity is a weighted average of the square of the times of the cash flows with weights

$$\left(\frac{C_t (1+i)^{-t}}{P(i)} \right)$$

Problem: A five-year zero-coupon bond redeemable at C is purchased to provide an annual effective yield of 6%. Find the modified convexity $C(i, 1)$. Use this, along with the modified duration $D(i, 1)$, to estimate the relative price change if the interest rate goes up by 100 basis points.

solution. $N=5$, $i=6\%$.

$$P(i) = C(1+i)^{-5} \quad P'(i) = -5C(1+i)^{-6}$$

$$P''(i) = 30C(1+i)^{-7}$$

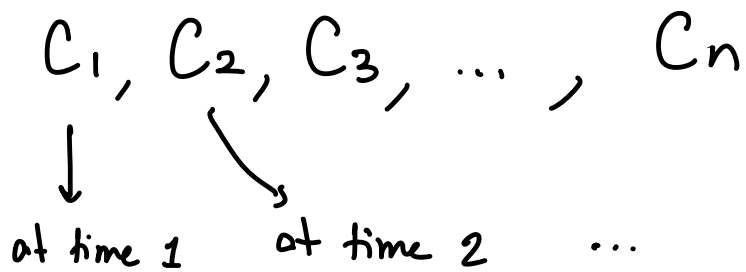
$$D(i, 1) = \frac{5}{(1+i)} \quad C(i, 1) = \frac{30}{(1+i)^2}$$

$$\frac{P(7\%) - P(6\%)}{P(6\%)} \approx -D(6\%, 1)(1\%) + C(6\%, 1) \frac{(1\%)^2}{2}$$

$$= -0.04583$$

Duration Recap!

Idea #1: Suppose we want to know a weighted average of the times of a series of cash flows



Natural weights: $w(t) = \frac{PV(\text{cash flow } C_t)}{PV(\text{All cash flows})}$

$$MacD = \sum_{t=1}^n t \cdot w(t)$$

MacD measures a weighted average of the time of a set of cash flows.

Idea #2: $P(i) = \sum_{t=1}^{\infty} C_t (1+i)^{-t}$ is a

decreasing function as i grows so $-P'(i)$ measures how fast this decline is. If we want to normalize this we get

$$\text{Mod } D = \frac{-P'(i)}{P(i)}$$

Now if we actually compute $-P'(i)$ we get

$$P(i) = C_1(1+i)^{-1} + C_2(1+i)^{-2} + C_3(1+i)^{-3} + \dots$$

$$-P'(i) = 1 \cdot C_1(1+i)^{-2} + 2C_2(1+i)^{-3} + 3C_3(1+i)^{-4} + \dots$$

FACTOR OUT A $(1+i)^{-1}$ term

$$-P'(i) = (1+i)^{-1} \left[1 \cdot C_1(1+i)^{-1} + 2 \cdot C_2(1+i)^{-2} + \dots \right]$$

$$\text{so } \frac{-P'(i)}{P(i)} = (1+i)^{-1} \left[\frac{1 \cdot C_1(1+i)^{-1}}{P(i)} + \frac{2C_2(1+i)^{-2}}{P(i)} + \dots \right]$$


1. $w(1)$

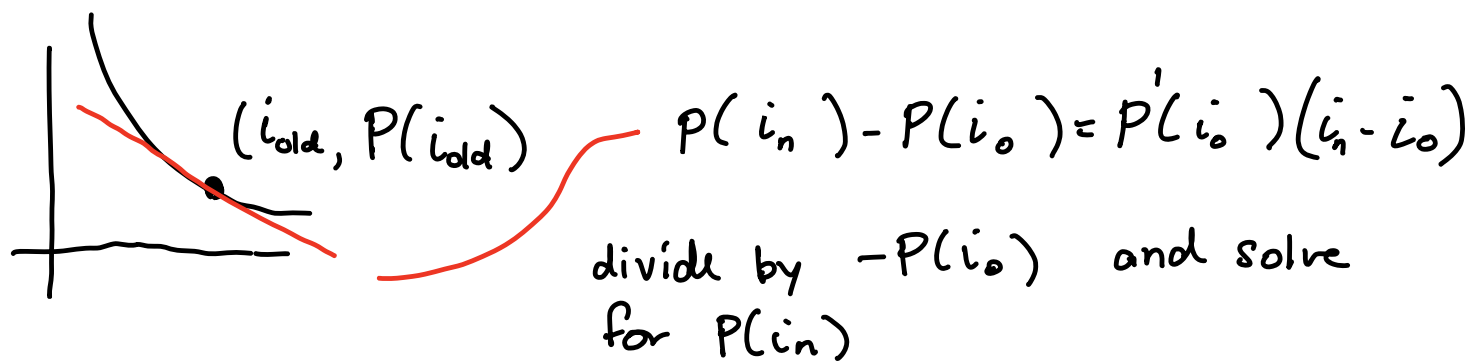
2. $w(2)$

so

$$\text{Mod } D = (1+i)^{-1} (\text{Mac } D)$$

It is not a coincidence that if we want to estimate a p.v. change when interest rates change, duration plays a role:

let $i_{\text{old}} \rightarrow i_{\text{new}}$ be some small interest rate change



$$P(i_n) \approx P(i_0) [1 - (i_n - i_0) \text{Mod } D]$$

modified duration 1st order approximation

Convexity Recap: It is not too difficult to see that given $P(i) = \sum_{t=1}^{\infty} C_t (1+i)^{-t}$, $P''(i) > 0$. Therefore we can find a normalized version, just like in Mod D.

$$\text{Mod } C = \frac{P''(i)}{P(i)}$$

If we actually compute $P''(i)$ we get

$$P(i) = C_1(1+i)^{-1} + C_2(1+i)^{-2} + C_3(1+i)^{-3} + \dots$$

$$P'(i) = -[C_1(1+i)^{-2} + 2C_2(1+i)^{-3} + 3C_3(1+i)^{-4} + \dots]$$

$$P''(i) = 2 \cdot 1 C_1(1+i)^{-3} + 3 \cdot 2 C_2(1+i)^{-4} + 4 \cdot 3 C_3(1+i)^{-5} + \dots$$

Again by factoring out $(1+i)^{-2}$ and normalizing

$$\frac{P''(i)}{P(i)} = (1+i)^{-2} \left[\sum_{t=1}^{\infty} \frac{t(t+1) C_t (1+i)^{-t}}{P(i)} \right]$$

$w(t)$

So $\text{Mod } C = (1+i)^{-2} \left[\sum_{t=1}^n t^2 w(t) + \sum_{t=1}^n t w(t) \right]$

Mac C Mac D

1st order Mac Dur Approx

$$P(i_n) = P(i_0) \left(\frac{1+i_0}{1+i_n} \right)^{\text{Mac D}}$$

$$P(s) = \sum_{t \geq 0} C_t e^{-st}$$

$$-P'(s) = \sum t C_t e^{-st}$$

Goal: Show MacD of an n -year
per value bond is $\ddot{a}_{\overline{n}|}$

$$\text{MacD} = \frac{\sum_{t=1}^n tw(t)}{\text{PV of bond}}$$

$$= \frac{\sum_{t=1}^n t Fr v^t + n Fv^n}{Fr \ddot{a}_{\overline{n}|} + Fv^n}$$


$$\rightarrow Fr [v + 2v^2 + 3v^3 + \dots + nv^n]$$

$$= Fr \left[\ddot{a}_{\overline{n}|} + \frac{1}{i} (\ddot{a}_{\overline{n}|} - nv^n) \right]$$

$$Fr \ddot{a}_{\overline{n}|} + Fv^n$$

$$= \frac{ra\overline{n} + \frac{r}{i}a\overline{n} - \frac{r}{i}nv^n}{ra\overline{n} + v^n}$$

$$= \frac{a\overline{n} + \frac{1}{i}a\overline{n} - \frac{1}{i}nv^n}{a\overline{n} + \frac{v^n}{r}}$$

$$= \frac{a\overline{n} \left(\frac{1+i}{i} \right) - \frac{1}{i}nv^n}{a\overline{n} + \frac{v^n}{r}}$$

$$= \frac{\frac{1}{i}(\ddot{a}\overline{n} - nv^n)}{a\overline{n} + \frac{v^n}{r}}$$